

Project 6 — AI/ML Dataset Guide

Smart Energy Grid & Solar Power Forecasting with Battery Optimization

This guide describes the six CSV datasets delivered to the AI/ML team. It explains the resolution, aggregation level, column structure, intended use, and recommended modeling path. The contents and counts are derived directly from the supplied DATASETS ZIP.

1. Dataset Package at a Glance

Dataset	Rows	Cols	Resolution	Level	Recommended use
plant1_ai_ml_handoff_hourly.csv	816	23	Hourly	Plant	PRIMARY Week-1 baseline
plant2_ai_ml_handoff_hourly.csv	816	23	Hourly	Plant	PRIMARY Week-1 baseline
plant1_ai_ml_handoff_15min_plantlevel.csv	3,158	27	15-min	Plant	Higher-frequency plant model
plant2_ai_ml_handoff_15min_plantlevel.csv	3,259	27	15-min	Plant	Higher-frequency plant model
plant1_ai_ml_handoff_15min.csv	68,778	23	15-min	Inverter	Advanced/inverter-level option
plant2_ai_ml_handoff_15min.csv	67,698	23	15-min	Inverter	Advanced/inverter-level option

Important clarification about column counts: The 15-minute plant-level datasets have **27 columns** because they retain additional plant aggregation and interval/energy-calculation fields. The 15-minute inverter datasets have **23 columns**, while the hourly plant-level datasets have **23 columns** because some intermediate fields are not retained after hourly aggregation. The different column counts are intentional.

2. Why Plant-Level Has 27 Columns

The 15-minute plant-level files aggregate all inverter readings at each timestamp and additionally retain intermediate fields needed to describe plant-wide energy and interval validity. Their 27 columns are:

Column group	Columns
Plant/time	time, plant_id
Plant generation	dc_power, ac_power, daily_yield, inverter_count
Interval/energy calculations	ac_dc_efficiency, next_interval_hours, interval_valid_for_energy, ac_energy_kwh_est, dc_energy_kwh_est
Plant sensor weather	sensor_temperature, module_temperature, sensor_irradiation, sensor_weather_interpolated
Open-Meteo weather/solar	openmeteo_temperature, humidity, cloud, cloud_low, cloud_mid, cloud_high, rain, ghi, direct_radiation, dhi, dni, wind_speed_10m

Total: 2 + 4 + 5 + 4 + 12 = 27 columns. These extra five interval/energy fields are the main reason the plant-level 15-minute dataset is wider than the 23-column datasets.

3. 23-Column Datasets — Why They Have 23

15-minute inverter-level (23 columns): 7 inverter/generation fields + 4 plant-sensor fields + 12 Open-Meteo fields. These rows preserve individual inverter observations and therefore repeated timestamps are expected.

Hourly plant-level (23 columns): hourly plant power/energy fields + plant weather + Open-Meteo features + an hourly completeness flag. Hourly aggregation intentionally produces a modeling-oriented schema rather than carrying every 15-minute intermediate field.

4. What Each Dataset Represents

Plant 1 — 15-min inverter-level — 68,778 rows × 23 columns

Full-resolution individual inverter observations with integrated plant weather and Open-Meteo features. **Recommended use:** Use for inverter-level modeling, inverter comparisons, anomaly/fault analysis, or advanced high-frequency experiments.

Plant 2 — 15-min inverter-level — 67,698 rows × 23 columns

Full-resolution individual inverter observations with integrated plant weather and Open-Meteo features. **Recommended use:** Same as Plant 1. Plant 2 also retains the known TOTAL_YIELD anomaly; do not use total_yield as the main target.

Plant 1 — 15-min plant-level — 3,158 rows × 27 columns

One plant-wide observation per generation timestamp after aggregating inverter readings. **Recommended use:** Best high-frequency option for total plant AC-power forecasting.

Plant 2 — 15-min plant-level — 3,259 rows × 27 columns

One plant-wide observation per generation timestamp after aggregating inverter readings. **Recommended use:** Best high-frequency option for total plant AC-power forecasting.

Plant 1 — hourly plant-level — 816 rows × 23 columns

Plant-wide solar generation and weather/Open-Meteo features aggregated to hourly resolution. **Recommended use:** Recommended primary Week-1 baseline.

Plant 2 — hourly plant-level — 816 rows × 23 columns

Plant-wide solar generation and weather/Open-Meteo features aggregated to hourly resolution. **Recommended use:** Recommended primary Week-1 baseline.

5. Key Column Groups

Group	Columns	Purpose
Time/identity	time, plant_id, source_key	Timestamp and plant/inverter identity. source_key is present in inverter-level data.
Generation	ac_power, dc_power, daily_yield, total_yield	Solar generation measurements. Use ac_power as the primary forecasting target.
Plant aggregation	inverter_count / avg_inverter_count	Number of inverter records represented in the plant observation.
Energy/interval	ac_energy_kwh_est, dc_energy_kwh_est, next_interval_hours, interval_valid_for_energy, ac_dc_efficiency	Plant-level 15-minute calculation fields; mainly useful for plant-level analysis.
Plant sensor weather	sensor_temperature, module_temperature, sensor_irradiation	On-site weather/irradiance measurements.
Open-Meteo weather	openmeteo_temperature, humidity, rain, wind_speed_10m	External weather features.
Open-Meteo solar/cloud	ghi, direct_radiation, dhi, dni, cloud, cloud_low, cloud_mid, cloud_high	Solar radiation and cloud features useful for forecasting.

6. Which Dataset Should AI/ML Use?

Goal	Recommended dataset	Reason
Week-1 baseline forecasting	Both hourly plant-level files	Clean plant-wide hourly time series; simplest starting point and aligns with baseline scope.
15-minute plant forecasting	Both 15-min plant-level files	Higher resolution while avoiding repeated inverter rows.
Inverter-level modeling	Both 15-min inverter-level files	Preserves every inverter observation and source_key.

7. Recommended Week-1 Modeling Workflow

- 1 Start with **plant1_ai_ml_handoff_hourly.csv** and **plant2_ai_ml_handoff_hourly.csv**.
- 2 Use **ac_power** as the primary forecasting target.
- 3 Use a chronological/time-based train/validation/test split; do not randomly shuffle the time series.
- 4 Engineer time features and the solar/weather features already supplied.
- 5 Add lag features (such as t-1 and t-24 where appropriate), rolling irradiance, and cloud-related features.
- 6 Handle nighttime carefully because zero solar generation is expected.
- 7 Evaluate the baseline using RMSE and MAPE with appropriate handling of zero/near-zero actual generation.
- 8 If higher-frequency modeling is desired, move to the 15-minute plant-level files before using inverter-level data.

8. Important Notes

- Repeated timestamps in inverter-level files are expected because multiple inverters report simultaneously.
- Plant 2 contains known anomalous TOTAL_YIELD values. They are preserved for traceability; use ac_power as the primary target.
- Do not duplicate/oversample records simply to increase dataset size.
- Weather and Open-Meteo features are already integrated into all six handoff datasets.
- The plant-level 15-minute files are richer than the 23-column files because they retain interval and energy calculation fields.

Final recommendation: Give the AI/ML team the DATASETS ZIP plus this guide. Let them choose the modeling resolution, with the hourly plant-level datasets as the recommended starting point.